PROJECT REQUIREMENTS

Automated Web Scanner: Secure Headers and CORS Flaws

1. Hardware Requirements

- Desktop or laptop with a modern 64-bit operating system.
- 4 GB RAM minimum; 8 GB recommended.
- Network connectivity to the approved test website(s).

2. Software Requirements

- Python 3.10 or newer.
- Windows, Linux, or macOS.
- Code editor or IDE such as VS Code, PyCharm, or a text editor.
- A test/staging website or written authorization for every target.

3. Python Dependencies

No third-party packages are required. The script uses Python standard-library modules for HTTP requests, URL parsing, JSON output, timestamps, and command-line arguments.

4. Functional Requirements

- Accept one or more HTTP/HTTPS URLs.
- Send ordinary GET and OPTIONS requests with a configurable timeout.
- Record response status codes and observed response headers.
- Check selected security headers and basic CORS response indicators.
- Print a concise console summary.
- Save structured results to JSON.

5. Security and Ethical Requirements

- Scan only sites owned by the operator or explicitly authorized for testing.
- Avoid intrusive exploitation, brute force, credential attempts, and link crawling.
- Keep the request rate low and respect site rules and scope.
- Do not include secrets, session tokens, or personal data in reports.
- Validate findings manually before assigning risk or changing configurations.

6. Validation Criteria

- Valid URLs produce a report with status codes and observed headers.
- Invalid URLs are reported without stopping valid targets from being processed.
- Missing/weak security headers are flagged for review.
- Basic CORS response behavior is described without claiming proof of exploitability.
- The JSON output is valid and suitable for later analysis.

Note: Header recommendations depend on application behavior, browser support, framework configuration, and organizational security standards.